GLOBE SCOTT MOTORS · FREE ENGINEERING RESOURCE

Motor Voltage Compatibility Checklist

For maintenance engineers and OEM designers matching DC motors to battery, rectifier and DC-bus supplies

Complete Steps 1 and 2 for every new installation or supply change. Steps 3 and 4 flag the risks of a mismatch and Step 5 selects the remedy. Measure voltages at the motor terminals under load as well as at the source.

Step 1 — Survey the supply voltage window

- [ ] Nominal supply or battery voltage: _____ V
- [ ] Minimum voltage at the motor terminals under full load (discharged battery or weak supply): _____ V
- [ ] Maximum voltage at the motor terminals (charging, float or light-load supply): _____ V
- [ ] Rectified AC supply: note whether it is filtered. Capacitor-filtered 230 V AC gives about 310–325 V DC; unfiltered full-wave averages about 207 V
- [ ] Voltage drop between source and motor terminals at rated current: _____ V (aim for no more than about 3 % of nominal)
- [ ] Regenerative braking: does the bus rise above the charge voltage during braking? Yes / No

Step 2 — Check the motor data

- [ ] Nameplate voltage, current, speed and power: _____
- [ ] Duty type (S1, S2 or S3) and insulation class: _____
- [ ] Measured no-load speed: _____ rpm at _____ V; Ke = (V - brush drop) / no-load speed in rad/s
- [ ] Armature resistance at 20 °C: _____ Ω; stall current = (V - brush drop) / Ra
- [ ] Permitted supply window from the data sheet or manufacturer: _____ V to _____ V

Step 3 — Overvoltage risk checks

- [ ] Speed ratio = maximum supply voltage / rated voltage; centrifugal and unbalance forces scale with its square
- [ ] Gearbox and driven-machine maximum input speed checked against the raised speed
- [ ] Stall current at maximum voltage within fuse, controller and wiring ratings
- [ ] Reversing current = (V + back-EMF - brush drop) / Ra; controller prevents reversal at full voltage and speed
- [ ] Fan, blower or centrifugal pump load: never apply overvoltage, as shaft power rises roughly with the cube of speed
- [ ] Commutator sparking and brush wear inspected early in service after any supply change

Step 4 — Undervoltage risk checks

- [ ] Speed at rated torque = (minimum V - I × Ra - brush drop) / Ke is still acceptable
- [ ] Stall torque at minimum voltage exceeds the load breakaway torque with margin
- [ ] Winding temperature checked at the lowest operating speed if the motor relies on a shaft-mounted fan
- [ ] Power-regulated drives: input current = power / (voltage × efficiency) at minimum voltage is within cable and controller limits
- [ ] Controller low-voltage cut-off set above the battery's minimum safe voltage

Step 5 — Choose the remedy

- [ ] Wrong bus: correct the supply, transformer or charger tap
- [ ] Bus higher than winding: DC-DC converter, or PWM controller with maximum duty = motor voltage / bus voltage and a current limit at motor rating
- [ ] Do not use series dropping resistors: high loss, and no voltage drop at light load
- [ ] Long term: specify or rewind for the bus voltage; turns scale with V2/V1 and resistance with $(V2/V1)^2$
- [ ] Re-test any rewound armature with surge, high-voltage and load tests before returning it to service

Typical DC Supply Voltage Windows

Supply	Nominal	Typical minimum	Typical maximum	Note
Lead-acid, 6 cells	12 V	10.5 V	14.4 V	1.75 V/cell discharged, 2.4 V/cell charging
Lead-acid, 12 cells	24 V	21.0 V	28.8 V	Equalising charge may go higher
Lead-acid, 24 cells	48 V	42.0 V	57.6 V	Close to the common 60 V DC threshold
LiFePO4, 4 cells	12.8 V	10.0 V	14.6 V	2.5–3.65 V/cell; BMS limits apply
LiFePO4, 8 cells	25.6 V	20.0 V	29.2 V	BMS may cut off earlier
LiFePO4, 16 cells	51.2 V	40.0 V	58.4 V	Check controller voltage rating
230 V AC, full-wave, unfiltered	About 207 V average	–	About 325 V peak	High ripple; use a drive
230 V AC, capacitor-filtered	About 310–325 V	–	–	Suits 310 V BLDC windings, not 180 V PMDC

Values are typical; confirm them against the battery maker's charge profile and the motor data sheet. The 60 V DC figure is a threshold used in many product and installation standards, not a universal rule, so check the standard that applies to your equipment.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.